Elliptic Integrals Built From Ratios Of Cubic Bézier Arc Lengths and Constructible Numbers

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Abstract

We show that E(arctan(ϕ),k) and F(arctan(ϕ),k) can be built from the ratio of the arc length of a section of a Cubic Bézier curve to the full length, defined as L(a) and numbers that can be constructed using a straightedge and a compass.

1 Basic Definitions

1. The Elliptic Integral of the First Kind, $F(\phi, k) = \int_0^\phi \frac{d\theta}{\sqrt{1-k^2\sin^2(\theta)}}$
2. The Elliptic Integral of the Second Kind, $E(\phi, k) = \int_0^\phi \sqrt{1-k^2\sin^2(\theta)}$
3. Delta Function, $\Delta(\theta, k) = \sqrt{1-k^2\sin^2(\theta)}$
4. Arc Length Function, $A(X(t), Y(t))_a^b = \int_a^b \sqrt{(\frac{d}{dt}X(t))^2 + (\frac{d}{dt}Y(t))^2}dt$
5. Ratio Arc Length Function, Defined as the ratio of the arc length of the curve to the full arc length from 0 to 1. It is defined as L(a) = $\frac{A(X(t),Y(t))_0^a}{A(X(t),Y(t)_0^1}$

2 Identities

1. $F(\theta, k)+F(\phi, k) = F(\psi, k)$ where $\psi = \arcsin(\frac{(\sin(\theta)\cos(\phi))\Delta(\phi,k)+(\sin(\phi)\cos(\theta))\Delta(\theta,k)}{1-k^2\sin^2(\theta)\sin^2(\phi)})$
2. $E(\theta, k) + E(\phi, k) = E(\psi, k) + k^2\sin(\theta)\sin(\phi)\sin(\psi)$

Using the two variables k_2 and ϕ_2 we can perform a Landen Transform

3. $k_2 = \frac{2\sqrt{k}}{1+k}$
4. $2\phi_2 = \phi + \arcsin(k\sin(\phi)$
5. $F(\phi, k) = \frac{2}{1+k}F(\phi_2, k_2)$,
6. $E(\phi, k) = (1+k)E(\phi_2, k_2) + (1-k)F(\phi_2, k_2) - k\sin(\phi)$

3 Proof and Construction

To construct our elliptic integrals we begin by considering the curve defined by the parametric definition of $X(t) = \frac{t}{3} - t$ and $Y(t) = t^2\sqrt{1-k^2}$ where k is the modulus of the integral we will be building.

Definition 1. *We define* $R_1(x) = \frac{2x\left(x^2+A+1\right)\sqrt{x^4+Ax^2+1}}{(1+x^2)}$
(A or a used interchangably)is a substitution made to simplify the expression and is equal to $4 - 2k^2$

Definition 2. *We define* $e_{xpr}(x) = \left(1-k^2\right)F\left(2\arctan\left(x\right),k\right) - \left(1-2k^2\right)E\left(2\arctan\left(x\right),k\right)$

Theorem 1. *For our Bézier curve defined above, its arc length from 0 to x is given by* $\frac{R_1(x)+4e_{xpr}(x)}{6}$

Proof. The arc length of our curve from 0 to x is given by $A(X(t), Y(t)$ or $\int_0^x \sqrt{(\frac{d}{dt}X(t))^2 + (\frac{d}{dt}Y(t))^2}dt$
When we consider our curve the arclength will be
$\int_0^x \sqrt{\left(\frac{d}{dt}\left(\frac{t^3}{3} - t\right)\right)^2 + \left(\frac{d}{dt}\left(t^2\sqrt{1-k^2}\right)\right)^2}dt$
We take the derivative of X(t) first, applying the power rule the derivative is $\frac{d}{dx}X(t) = t^2 - 1$
Then we take the derivative of Y(t), taking the constant outside the derivative and applying the power rule gives us $\frac{d}{dy}Y(t) = 2t\sqrt{1-k^2}$
This transforms our integral into $\int_0^x \sqrt{\left(t^2-1\right)^2 + \left(2t\sqrt{1-k^2}\right)^2}dt$
Next we square and add the terms $(t^2-1)^2 = t^4 - 2t^2 + 1$
$(2t\sqrt{1-k^2})^2 = 4t^2(1-k^2)$
$t^4 - 2t^2 + 1 + 4t^2(1-k^2) = t^4 + 2t^2 + 1 - 4t^2k^2$
Our integral can now be written as
$\int_0^x \sqrt{t^4 + t^2\left(2-4k^2\right) + 1}dt$
Since we claimed that this equals, $\frac{R_1(x)+4e_{xpr}(x)}{6}$
we can take take the derivative of both functions to prove our Lemma by the Fundamental Theorem of Calculus
The derivative of $\int_0^x \sqrt{t^4 + t^2\left(2-4k^2\right) + 1}dt$ is $\sqrt{x^4 + t^2\left(2-4k^2\right) + 1}$ by the Fundamental Theorem of Calculus
We can substitute $A = 2 - 4k^2$ turning our integral into $\sqrt{x^4 + Ax^2 + 1}$

For the derivative of $\frac{R_1(x)+4e_{xpr}(x)}{6}$ we split it into $\frac{\frac{d}{dx}(R_1(x))+4\frac{d}{dx}(e_{xpr}(x))}{6}$

The derivative of $R_1(x)$ can be computed by first. We start by taking the ln of both sides of $y = \frac{2x\left(x^2+A+1\right)\sqrt{x^4+Ax^2+1}}{(1+x^2)}$
This becomes $\ln(y) = \ln\left(x^2+A+1\right) + \frac{1}{2}\ln\left(x^4+Ax^2+1\right) - \ln\left(1+x^2\right) + \ln(x) + \ln(2)$
Now we can differentiate $\frac{d}{dx}\ln(y) = \frac{d}{dx}\ln\left(x^2+A+1\right) + \frac{d}{dx}\frac{1}{2}\ln\left(x^4+Ax^2+1\right) - \frac{d}{dx}\ln\left(1+x^2\right) + \frac{d}{dx}\ln(x) + \frac{d}{dx}\ln(2)$ The derivative of the left side is $\frac{1}{y}\frac{dy}{dx}$
For the derivative of the right side, $\frac{d}{dx}\ln(2) = 0$, $\frac{d}{dx}\ln(x) = \frac{1}{x}$, $\frac{d}{dx}\ln\left(1+x^2\right) = \frac{2x}{1+x^2}$ by the chain rule, $\frac{d}{dx}\frac{1}{2}\ln\left(x^4+Ax^2+1\right) = \frac{1}{2}\frac{d}{dx}\ln\left(x^4+Ax^2+1\right) = \frac{2x^3+Ax}{x^4+Ax^2+1}$, $\frac{d}{dx}\ln\left(x^2+A+1\right) = \frac{2x}{x^2+A+1}$
Finally we multiply by y to solve for the derivative and substitute in y from earlier.
$\frac{dy}{dx} = y\frac{2x}{x^2+Ax+1} + \frac{2x^3+Ax}{x^4+Ax^2+1} - \frac{2x}{1+x^2} + \frac{1}{x}$
$\frac{dy}{dx} = \frac{2x\left(x^2+A+1\right)\sqrt{x^4+Ax^2+1}}{(1+x^2)}\left(\frac{2x}{x^2+A+1} + \frac{2x^3+Ax}{x^4+Ax^2+1} - \frac{2x}{1+x^2} + \frac{1}{x}\right)$

Now we differentiate $4e_{xpr}(x)$, we can pull out the constant to get $4\frac{d}{dx}e_{xpr}(x)$
Now to differentiate $e_{xpr}(x)$ we can split it, $\frac{d}{dx}e_{xpr}(x) = \frac{d}{dx}\left(1-k^2\right)F\left(2\arctan(x),k\right) - \frac{d}{dx}\left(1-2k^2\right)E\left(2\arctan(x),k\right)$
We can also remove the constants $\left(1-k^2\right)\frac{d}{dx}F\left(2\arctan(x),k\right) - \left(1-2k^2\right)\frac{d}{dx}E\left(2\arctan(x),k\right)$
We start by differentiating $F\left(2\arctan(x),k\right)$
Lets bring in our definition for F, $F(\phi,k) = \int_0^{\phi}\frac{d\theta}{\sqrt{1-k^2\sin^2(\theta)}}$
Our derivative becomes $\frac{d}{dx}\int_0^{2\arctan(x)}\frac{1}{\sqrt{1-k^2\sin^2\theta}}dt$ We get $\frac{2}{1+x^2}\cdot\frac{1}{\sqrt{1-k^2\sin^2(2\arctan(x))}}$ by the Leibniz Rule and $\frac{d}{dx}\arctan(x) = \frac{1}{1+x^2}$
Simplifying $\sin^2\left(2\arctan(x)\right) = \frac{4x^2}{(1+x^2)^2}$ we get $\frac{2}{1+x^2}\cdot\frac{1}{\sqrt{1-k^2\frac{4x^2}{(1+x^2)^2}}}$ then $\frac{2}{\sqrt{(1+x^2)^2\left(1-k^2\frac{4x^2}{(1+x^2)^2}\right)}}$ by bringing the $\frac{1}{1+x^2}$ inside the radical. Multiplying and taking the square root we get $\frac{2}{\sqrt{x^4+(2-4k^2)x^2+1}}$.
Next we take the derivative of $E\left(2\arctan(x),k\right)$, again we bring in the definiton.
$E(\phi,k) = \int_0^{\phi}\sqrt{1-k^2\sin^2(\theta)}$
Again we apply Leinbniz rule $\frac{d}{dx}\int_0^{2\arctan(x)}\sqrt{1-k^2\sin^2(\theta)}d\theta = \sqrt{1-\frac{4k^2x^2}{(1+x^2)^2}}\cdot\frac{2}{1+x^2}$
Then simplify, $\frac{2\sqrt{(1+x^2)^2-4k^2x^2}}{(1+x^2)^2}$

Next we multiply by our constants and sum $\frac{\left(1-k^2\right)2}{\sqrt{x^4+(2-4k^2)x^2+1}}-\frac{\left(1-2k^2\right)2\sqrt{(1+x^2)^2-4k^2x^2}}{\left(1+x^2\right)^2}$
Now we simplify the whole expression $\frac{\left(2-2k^2\right)\left(x^2+1\right)^2-A\left(x^4+Ax^2+1\right)}{(x^2+1)^2\sqrt{x^4+Ax^2+1}}$
Now that we have taken the derivative of $R_1(x)$ and $e_{xpr}(x)$ we can compute the final expression. $\frac{\left(\frac{2x\left(x^2+A+1\right)\sqrt{x^4+Ax^2+1}}{\left(x^2+1\right)}\left(\frac{2x}{x^2+A+1}+\frac{2x^3+Ax}{x^4+Ax^2+1}-\frac{2x}{1+x^2}+\frac{1}{x}\right)+4\frac{\left(2-2k^2\right)\left(x^2+1\right)^2-A\left(x^4+Ax^2+1\right)}{(x^2+1)^2\sqrt{x^4+Ax^2+1}}\right)}{6}$
We simplify this expression and it turns out to equal $\sqrt{x^4+Ax^2+1}$
Because the derivative of our arc length equals the derivative of $\frac{R_1(x)+4e_{xpr}(x)}{6}$
Our arc length must equal it. Therefore our arc length equals $\frac{R_1(x)+4e_{xpr}(x)}{6}$
□

Definition 3. $h(x)=\frac{2x}{1-x^4}\sqrt{x^4+ax^2+1}$

Definition 4. *We define a another R function for arc length* $R_2(x)=3\left(\frac{h(x)\left(h(x)^2+a+1\right)\sqrt{h(x)^4+ah(x)^2+1}}{3\left(1+h(x)^2\right)}+\frac{a}{3}\frac{(2-a)}{4}\left(\frac{2x}{1+x^2}\right)^2\left(\frac{2h(x)}{1+h(x)^2}\right)\right)$

Lemma 1. *Our arc length of our curve from 0 to h(x) is* $\frac{4e_{xpr}(x)+R_2(x)}{3}$

Proof. We proved previously that our arc length is $\frac{4e_{xpr}(x)+R_1(x)}{6}$, now we plug in h(x), we get $\frac{4e_{xpr}\left(\frac{2x}{1-x^4}\sqrt{x^4+Ax^2+1}\right)+R_1\left(\frac{2x}{1-x^4}\sqrt{x^4+Ax^2+1}\right)}{6}$ First we evaluate $R_1(h(x))$
Next we evaluate $e_{xpr}\left(h\left(x\right)\right)$ this gives $\left(1-k^2\right)F\left(2\arctan\left(\frac{2x}{1-x^4}\sqrt{x^4+ax^2+1}\right),k\right)-\left(1-2k^2\right)E\left(2\arctan\left(\frac{2x}{1-x^4}\sqrt{x^4+ax^2+1}\right),k\right)$
Now we can use the addition identities for our integrals. In particular $F\left(\arcsin\left(\frac{\sin(2\phi)d(k,\phi)}{1-k^2\sin^2(\phi)\sin^2(\phi)}\right),k\right)=2F\left(\phi,k\right)$ First taking a look at our $F\left(2\arctan\left(\frac{2x}{1-x^4}\sqrt{x^4+ax^2+1}\right),k\right)$ term we can convert our arctanto an arcsin using $\arctan\left(x\right)=\arcsin\left(\frac{x}{\sqrt{1+x^2}}\right)$ now we convert we get $F\left(2\arcsin\left(\frac{\frac{2x}{1-x^4}\sqrt{x^4+ax^2+1}}{\sqrt{1+\left(\frac{2x}{1-x^4}\sqrt{x^4+ax^2+1}\right)^2}}\right),k\right)$
Then we use $2\arcsin\left(x\right)=\arcsin\left(2x\sqrt{1-x^2}\right)$ to get
$F\left(\arcsin\left(2\frac{\frac{2x}{1-x^4}\sqrt{x^4+ax^2+1}}{\sqrt{1+\left(\frac{2x}{1-x^4}\sqrt{x^4+ax^2+1}\right)^2}}\sqrt{1-\frac{\frac{2x}{1-x^4}\sqrt{x^4+ax^2+1}}{\sqrt{1+\left(\frac{2x}{1-x^4}\sqrt{x^4+ax^2+1}\right)^2}}^2}\right),k\right)$
Simplifying the inside we get $F\left(\arcsin\left(\frac{4x\left(1-x^4\right)\sqrt{x^4+ax^2+1}}{x^8+4x^6+(4a-2)x^4+4x^2+1}\right),k\right)$
This happens to be the exact form of $2F(\phi,k)$, where we have the phi =

$2\arctan(x)$ when we bring the 2 outside of our F we get $2F\left(2\arctan\left(x\right),k\right)$
Now we take a look at our E term, we can do something similar, $E\left(\arcsin\left(\frac{4x\left(1-x^4\right)\sqrt{x^4+ax^2+1}}{x^8+4x^6+(4a-2)x^4+4x^2+1}\right),k\right)$ this is our E term. Lets take a look at the similar duplication identity $2E\left(x,k\right)=E\left(\arcsin\left(\frac{\sin(2x)\sqrt{1-k^2\sin(x)^2}}{1-k^2(\sin(x))^4}\right),k\right)+k^2\sin\left(x\right)^2\frac{\sin(2x)\sqrt{1-k^2\sin(x)^2}}{1-k^2(\sin(x))^4}$
So now we can still do the trick using $2\arctan(x)$ but we have to subtract off the extra part $2E\left(2\arctan\left(x\right),k\right)-\frac{4(2-a)x^3\left(1-x^2\right)\sqrt{x^4+ax^2+1}}{(1+x^2)(x^8+4x^6+(4a-2)x^4+4x^2+1)}$
We now have this for $e_{xpr}(h(x)$,
$\left(1-k^2\right)2F\left(2\arctan\left(x\right),k\right)-a\left(E\left(2\arctan\left(x\right),k\right)-\frac{2(2-a)x^3\left(1-x^2\right)\sqrt{x^4+ax^2+1}}{(1+x^2)(x^8+4x^6+(4a-2)x^4+4x^2+1)}\right)$
Now we multiply by 4 and add this to $R_1(h(x)$ this gives us a complicated expression containing $h(x)$. However we can simplify this by combining the non elliptic and elliptic terms, and by factoring out a 2. Lets start with our elliptic term $\frac{\left(4\left(1-k^2\right)F(2\arctan(x),k)-2aE(2\arctan(x),k)\right)}{3}$
We can substitute in the definition for a and expand $\frac{4\left(1-k^2\right)F(2\arctan(x),k)-4\left(1-2k^2\right)E(2\arctan(x),k)}{3}$
Then factor out a 4, $\left(1-k^2\right)F\left(2\arctan\left(x\right),k\right)-\left(1-2k^2\right)E\left(2\arctan\left(x\right),k\right)$ is $e_{xpr}(x)$ so lets substitute that in and we get $\frac{4e_{xpr}(x)}{3}$, this indeed matches our claim. Now we need to show that the non-elliptic part is equal to $\frac{R_2(x)}{3}$.
We can substitute in h(x) in a few places in our non elliptic part to get that our non elliptic part is equal to $\frac{(h(x))\left(h(x)^2+a+1\right)\sqrt{h(x)^4+ah(x)^2+1}}{3\left(1+h(x)^2\right)}+2a\frac{2(2-a)x^3\left(1-x^2\right)\sqrt{x^4+ax^2+1}}{3(1+x^2)(x^8+4x^6+(4a-2)x^4+4x^2+1)}$
Our goal is to prove this is equal to $\frac{h(x)\left(h(x)^2+a+1\right)\sqrt{h(x)^4+ah(x)^2+1}}{3\left(1+h(x)^2\right)}+\frac{a}{3}\frac{(2-a)}{4}\left(\frac{2x}{1+x^2}\right)^2\left(\frac{2h(x)}{1+h(x)^2}\right)$ so we already have the left half done
Now we just need to prove $2a\frac{2(2-a)x^3\left(1-x^2\right)\sqrt{x^4+ax^2+1}}{3(1+x^2)(x^8+4x^6+(4a-2)x^4+4x^2+1)}=\frac{a}{3}\frac{(2-a)}{4}\left(\frac{2x}{1+x^2}\right)^2\left(\frac{2h(x)}{1+h(x)^2}\right)$
Factor out $\frac{a}{3}$ to get $\frac{4(2-a)x^3\left(1-x^2\right)\sqrt{x^4+ax^2+1}}{(1+x^2)(x^8+4x^6+(4a-2)x^4+4x^2+1)}$ now we factor out $\frac{(2-a)}{4}$ by multiplying numerator and denominator by 4 to get $\frac{a}{3}\frac{(2-a)}{4}\frac{16x^3\left(1-x^2\right)\sqrt{x^4+ax^2+1}}{(1+x^2)(x^8+4x^6+(4a-2)x^4+4x^2+1)}$ Now we multiply the fraction by $\frac{1+x^2}{1+x^2}$ to get $\frac{a}{3}\frac{(2-a)}{4}\frac{16x^3\left(1-x^2\right)\left(1+x^2\right)\sqrt{x^4+ax^2+1}}{(1+x^2)^2(x^8+4x^6+(4a-2)x^4+4x^2+1)}$, now we factor out $\left(\frac{2x}{1+x^2}\right)^2$ to get $\frac{a}{3}\frac{(2-a)}{4}\left(\frac{2x}{1+x^2}\right)^2\frac{4x\left(1-x^4\right)\sqrt{x^4+ax^2+1}}{(x^8+4x^6+(4a-2)x^4+4x^2+1)}$. Finally we show that $\frac{4x\left(1-x^4\right)\sqrt{x^4+ax^2+1}}{(x^8+4x^6+(4a-2)x^4+4x^2+1)}=\left(\frac{2h(x)}{1+h(x)^2}\right)$
$\frac{2h(x)}{1+h(x)^2}=\frac{4x\sqrt{x^4+ax^2+1}}{(1-x^4)\left(1+h(x)^2\right)}$

We factor out $4x\sqrt{x^4+ax^2+1}$ to get $\frac{\left(1-x^4\right)}{(x^8+4x^6+(4a-2)x^4+4x^2+1)}$ from both sides to get that we are proving $\frac{1}{(1-x^4)\left(1+h(x)^2\right)} = \frac{\left(1-x^4\right)}{(x^8+4x^6+(4a-2)x^4+4x^2+1)}$ then $\frac{1}{(1-x^4)\left(1+h(x)^2\right)} = \frac{1}{(1-x^4)\left(\frac{4x^2}{(1-x^4)^2}(x^4+ax^2+1)+1\right)}$

Simplify to $\frac{1}{\left(\frac{4x^2}{(1-x^4)}(x^4+ax^2+1)+(1-x^4)\right)}$ then $\frac{\left(1-x^4\right)}{\left(4x^2(x^4+ax^2+1)+(1-x^4)^2\right)}$. Finally, $4x^2\left(x^4+ax^2+1\right)+\left(1-x^4\right)^2 = x^8+4x^6+(4a-2)x^4+4x^2+1$ Which means $\frac{4x\left(1-x^4\right)\sqrt{x^4+ax^2+1}}{(x^8+4x^6+(4a-2)x^4+4x^2+1)}$ does indeed equal $\left(\frac{2h(x)}{1+h(x)^2}\right)$ which proves our non elliptic term is equal to $\frac{R_2(x)}{3}$ and finally proving that our total arc length is what we claimed. □

Theorem 2. *$e_{xpr}(x)$ can be constructed from the ratio function of our stated Bézier Curve $L(a)$*

Proof. We will prove our statement by directly constructing $e_{xpr}(x)$ from values of L(x) Our construction is very simple, lets start by considering L for our specific curve, $L(x) = \frac{4e_{xpr}(x)+R_1(x)}{4e_{xpr}(1)+R_1(1)}$
Now what we can do is take the ratio $\frac{L(h(x))}{L(x)}$, this causes the bottom terms to cancel giving us $\frac{8e_{xpr}(x)+2R_2(x)}{R_1(x)+4e_{xpr}(x)}$ then we subtract 2 giving us $\frac{2R_2(x)-2R_1(x)}{4e_{xpr}(x)+R_1(x)}$ then divide by 2, Now we take the reciprocal giving us $\frac{4e_{xpr}(x)+R_1(x)}{(R_2(x)-R_1(x))}$ Since $R_2(x)$ and $R_1(x)$ are easily constructible we multiply by $(R_2(x)-R_1(x))$ giving us $4e_{xpr}(x)+R_1(x)$ then we simply subtract $R_1(x)$ and divide by 4 proving we can construct $e_{xpr}(x)$ □

Theorem 3. *The Incomplete Elliptic integrals of the First and Second are constructible from $e_{xpr}(x)$*

Proof. We begin by preforming the Landen Transform on our Elliptic Integrals, we transform $F(\arctan(x),k)$ into $\frac{2}{1+k}F\left(\frac{\left(\arctan(x)+\arcsin\left(\frac{kx}{\sqrt{1+x^2}}\right)\right)}{2}, \frac{2\sqrt{k}}{1+k}\right)$ we can define $F_1(x) = F\left(\frac{\left(\arctan(x)+\arcsin\left(\frac{kx}{\sqrt{1+x^2}}\right)\right)}{2}, \frac{2\sqrt{k}}{1+k}\right)$ as a useful substitution and do the same for $E_1(x) = E\left(\frac{\left(\arctan(x)+\arcsin\left(\frac{kx}{\sqrt{1+x^2}}\right)\right)}{2}, \frac{2\sqrt{k}}{1+k}\right)$ then we also transform $E(\arctan x,k) = (1+k)E_1(x)+(1-k)F_1(x)-$

$\frac{kx}{\sqrt{1+x^2}}$
Now we take $e_{xpr}(x)$ in this new form with the transformed integrals and we get
$e_{xpr}(x) = \left(1-k^2\right)\frac{2}{1+k}F_1(x)+\left(1-2k^2\right)\left((1+k)E_1(x)+(1-k)F_1(x)-\frac{kx}{\sqrt{1+x^2}}\right)$
Now we can add $\left(\frac{kx}{\sqrt{1+x^2}}\cdot\left(1-2k^2\right)\right)$ to remove our non elliptic term and lets call the resulting function $f(x) = e_{xpr}(x)+\frac{\left(k-2k^3\right)x}{\sqrt{1+x^2}} = \left(1-k^2\right)\frac{2}{1+k}F_1(x)+\left(1-2k^2\right)((1+k)E_1(x)+(1-k)F_1(x))$
Now we expand this into $f(x) = \left(1+k-2k^2-2k^3\right)E_1(x)+\left(3-3k-2k^2+2k^3\right)F_1(x)$
We now build another function by evaluating $e_{xpr}(x)$ at the values for our transformed integral. We evaluate $e_{xpr}(\phi_2(x), k_2)$ giving us
$\left(1-\frac{2\sqrt{k}}{1+k}^2\right)F\left(\frac{\left(\arctan(x)+\arcsin\left(\frac{kx}{\sqrt{1+x^2}}\right)\right)}{2},\frac{2\sqrt{k}}{1+k}\right)+\left(1-2\frac{2\sqrt{k}}{1+k}^2\right)E\left(\frac{\left(\arctan(x)+\arcsin\left(\frac{kx}{\sqrt{1+x^2}}\right)\right)}{2},\frac{2\sqrt{k}}{1+k}\right)$
We can simplify this using our Landen Transform into $\frac{(1-k)^2F_1(x)+\left(k^2-6k+1\right)E_1(x)}{(1+k)^2}$
finally we multiply by $(1+k)^2$ to arrive at $g(x) = (1+k)^2 e_{xpr}(x_2(x), k_2) = (1-k)^2F_1(x)+\left(k^2-6k+1\right)E_1(x)$
We now have 2 functions f and g that we can set up a system of equations to solve for $E_1(x)$ and $F_1(x)$
We solve this system giving us $E_1(x) = \frac{\left(3-3k-2k^2+2k^3\right)g(x)-(1-k)^2f(x)}{2(2k^5-8k^4+4k^3+11k^2-10k+1)}$ and $F_1(x) = \frac{\left(k^2-6k+1\right)f(x)-\left(1+k-2k^2-2k^3\right)g(x)}{2(2k^5-8k^4+4k^3+11k^2-10k+1)}$
Finally, $F(\arctan(x), k) = \frac{2}{1+k}F_1(x)$ and $E(\arctan x, k) = (1+k)E_1(x)+(1-k)F_1(x)-\frac{kx}{\sqrt{1+x^2}}$
Because $f(x)$ and $g(x)$ are built from $e_{xpr}(x)$ which is built from our Bézier arc lenghts, the integrals are too. □

Theorem 4. *The complete elliptic integrals of the first and second kind can be built from our Bézier arc lengths.*

Proof. Since our integrals have $arctan(\phi)$ as the argument, we cant get $\frac{\pi}{2}$ as the argument. We have to do some transformations to obtain them. Using our duplication identities we can solve this issue. We want $\arcsin\left(\frac{\sin(2\phi)\Delta(\phi,k)}{1-k^2\sin^2(\phi)\sin^2(\phi)}\right) = \frac{\pi}{2}$ Lets solve for ϕ in terms of k
$\arcsin(1) = \frac{\pi}{2}$ so $\frac{\sin(2\phi)\Delta(\phi,k)}{1-k^2\sin(\phi)^4} = 1$ then
$\sin(2\phi)\sqrt{1-k^2\sin^2\phi} = 1-k^2\sin(\phi)^4$ We can substitute $p = sin^2(\phi)$ and square to get $\left(1-k^2p^2\right)^2 = 4p(1-p)$, Solving and undoing the substitution, and finally transforming

$\arcsin(x) = \arctan(\frac{x}{\sqrt{1-x^2}})$ we get $\phi = \arctan\left(\frac{1}{(1-k^2)^{\frac{1}{4}}}\right)$, now $F\left(\frac{\pi}{2}, x\right) = 2F\left(\arctan\left(\frac{1}{(1-x^2)^{\frac{1}{4}}}\right), x\right)$ and we add the extra term for the Second Kind to get
$2E\left(\arctan\left(\frac{1}{(1-x^2)^{\frac{1}{4}}}\right), x\right) - \frac{x^2}{1+\sqrt{1-x^2}} = E\left(\frac{\pi}{2}, x\right)$ This means the complete elliptic integrals are constructible from duplications of the incomplete ones. □